

Numerical Methods in Soft Matter Exercises

Guglielmo Bordin

University of Padua, 2024

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Sampling via hit-and-miss and inversion

1.1 Sampling in d -dimensional domains by hit-and-miss

Source codes `src/011a_rect_hit_miss.c, src/011b_disk_hit_miss.c`

Analysis scripts `src/011_hit_miss.R`

Executables `exe/011a_rect_hit_miss, exe/011b_disk_hit_miss`

The error as a function of the number of throws is shown in Fig. 1.1.1. In the disk case, it goes under 1 % with as little as 10 000 ‘throws’.

1.2 Sampling from a distribution with the inversion method

Source codes `src/012_inversion_power34.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/012_inversion_power34`

The idea is to sample from the probability distribution $\rho_n(x) = cx^n$ in $[0, 1]$. First, using the normalization condition we can find out what c should be:

$$1 = \int_0^1 cx^n dx = \frac{c}{n+1} \implies c = n+1. \quad (1.2.1)$$

Then, we find the expression of the associated cumulative density function:

$$F_n(x) = (n+1) \int_0^x y^n dy = x^{n+1}, \quad (1.2.2)$$

and invert it:

$$u = x^{n+1} \implies x = u^{1/(n+1)}. \quad (1.2.3)$$

So, we just have to sample a random floating point number between 0 and 1 and raise it to the power of $1/(n+1)$ to get $x \sim (n+1)x^n$. Results from a sampling of 50 000 iterations for $n = 3, 4$ are shown in Fig. 1.2.1.

1.3 Another sampling with the inversion method

Source codes `src/012_inversion_power2.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/012_inversion_power2`

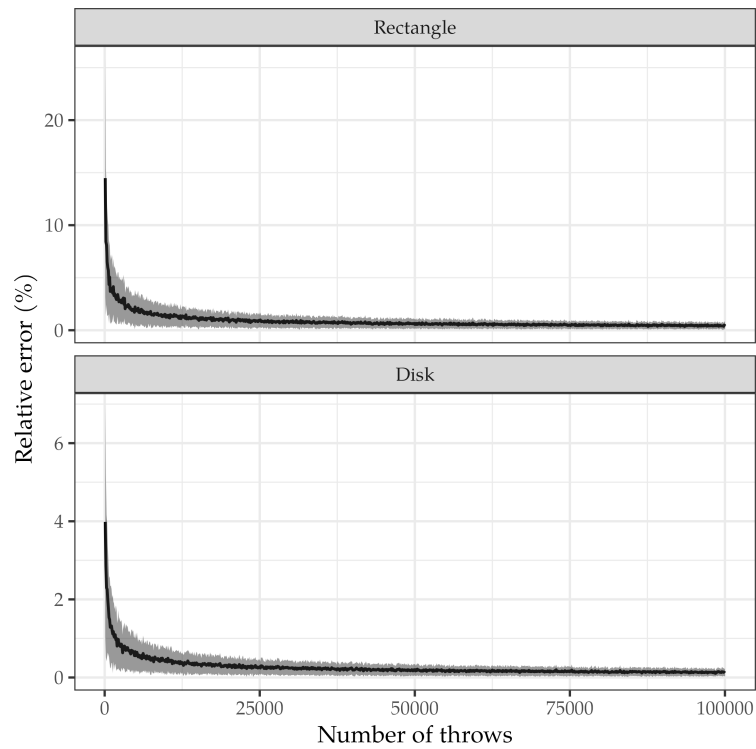


Figure 1.1.1: percentage error as a function of the number of ‘throws’ for Ex. 1.1. The black line represents the average over 100 realizations, with the standard deviation shaded in grey.

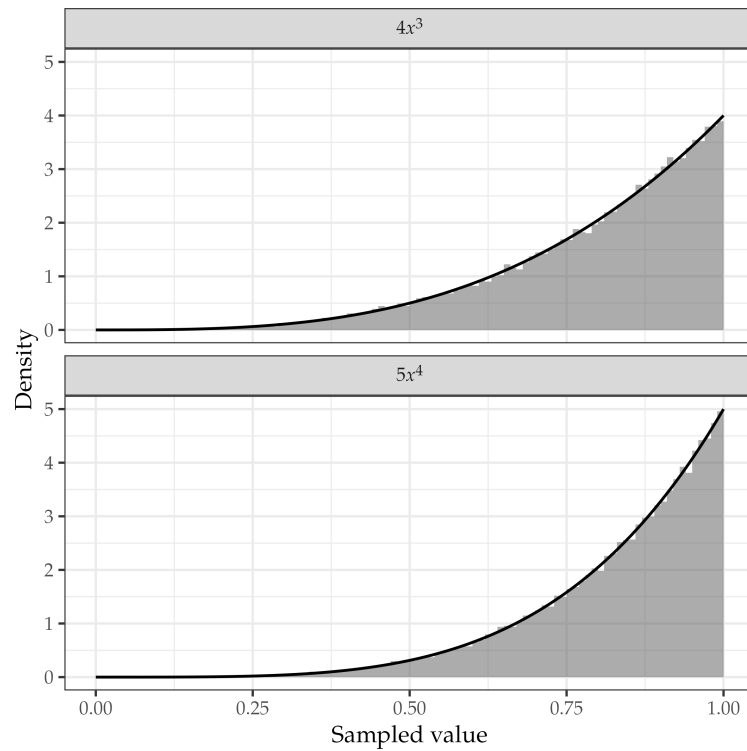


Figure 1.2.1: histogram of 50 000 samples from $\rho_n(x) = cx^n$, $x \in [0, 1]$, for $n = 3$ (above) and $n = 4$ (below), obtained with the inversion method. The theoretical distribution is overlaid with a solid line.

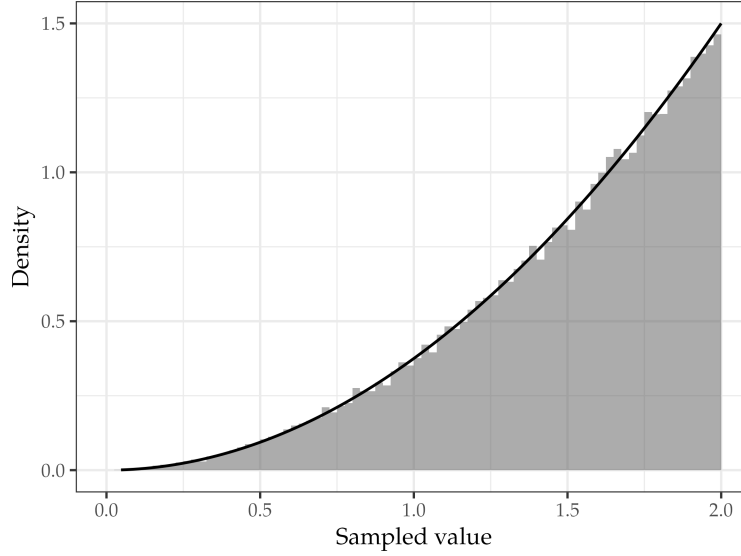


Figure 1.3.1: histogram of 50 000 samples from $\rho(x) = 3x^2/8$, $x \in [0, 2]$, obtained with the inversion method. The theoretical distribution is overlaid with a solid line.

I slightly modified the code from the previous exercise, since c is different now. The distribution is $\rho(x) = cx^2$ with $x \in [0, 2]$, so:

$$1 = \int_0^2 cx^2 dx = \frac{8}{3}c \implies c = \frac{3}{8}. \quad (1.3.1)$$

The cumulative is then

$$F(x) = \frac{3}{8} \int_0^x y^2 dy = \frac{x^3}{8} \implies x = 2u^{1/3}. \quad (1.3.2)$$

Once again, you can see the comparison between a 50 000-points histogram and the theoretical curve in Fig. 1.3.1.

1.4 Additional exercises on the inversion method

Source codes `src/014a_inversion_exp.c`, `src/014b_inversion_exp2.c`,
 `src/014c_inversion_powerlaw.c`

Analysis scripts `src/012-4_inversion_method.R`

Executables `exe/014_inversion_exp`, `exe/014b_inversion_exp2`,
 `exe/014c_inversion_powerlaw`

The first exercise here asked to sample from $\rho_1(x) = \mu e^{-\mu x}$, with $x \in [0, \infty)$. The cumulative and the sampling rule are then

$$F_1(x) = \mu \int_0^x e^{-\mu y} dy = \mu(1 - e^{-\mu x}) \implies x = -\log(1 - u)/\mu. \quad (1.4.1)$$

I ran 50 000 samples as before, choosing $\mu = 2$; see Fig. 1.4.1.

The second distribution was $\rho_2(x) = 2xe^{-x^2}$, $x \in [0, \infty)$. Cumulative and sampling rule:

$$F_2(x) = 2 \int_0^x e^{-y^2} dy = 1 - e^{-x^2} \implies x = \sqrt{-\log(1 - u)}. \quad (1.4.2)$$

A histogram of 50 000 samples is shown in Fig. 1.4.2.

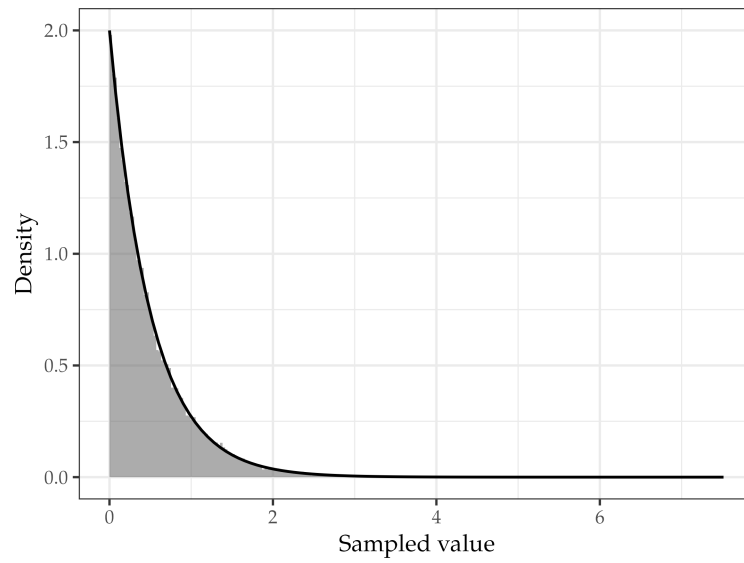


Figure 1.4.1: histogram of 50 000 samples from $\rho(x) = 2e^{-2x}$, $x \in [0, \infty)$, obtained with the inversion method. The theoretical distribution is overlaid with a solid line.

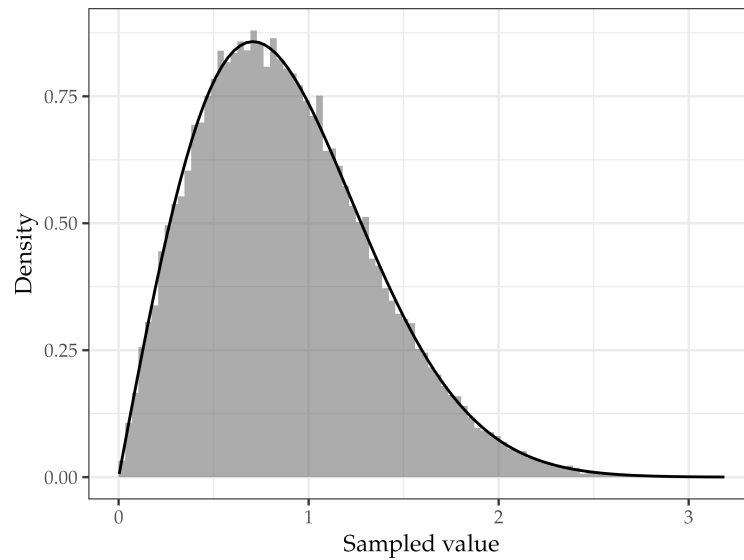


Figure 1.4.2: histogram of 50 000 samples from $\rho(x) = 2xe^{-x^2}$, $x \in [0, \infty)$, obtained with the inversion method. The theoretical distribution is overlaid with a solid line.

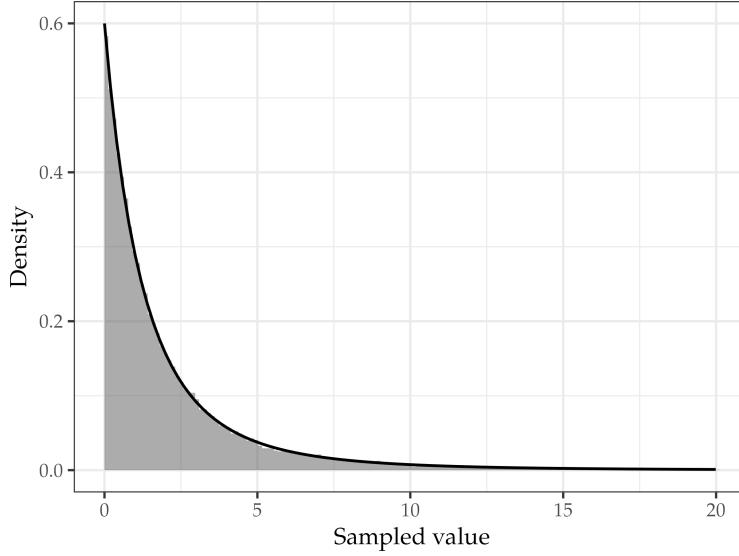


Figure 1.4.3: histogram of 50 000 samples from $\rho(x) = 375/(5+x)^4$, $x \in [0, \infty)$, obtained with the inversion method. The theoretical distribution is overlaid with a solid line.

The last distribution was $\rho_3(x) = c(a+bx)^{-n}$, with $x \in [0, \infty)$ and $n > 1$. The normalization constant is given by

$$1 = \int_0^\infty c(a+bx)^{-n} dx = \frac{a^{1-n}c}{b(n-1)} \implies c = b(n-1)a^{n-1}. \quad (1.4.3)$$

By integrating we obtain

$$F_3(x) = c \int_0^x (a+by)^{-n} dy = \frac{c}{b(1-n)} [(a+bx)^{1-n} - a^{1-n}] = 1 - a^{n-1}(a+bx)^{1-n}, \quad (1.4.4)$$

from which we can sample with

$$u = 1 - a^{n-1}(a+bx)^{1-n} \implies x = \frac{a}{b} [(1-u)^{1/(1-n)} - 1]. \quad (1.4.5)$$

I chose $a = 1/2$, $b = 1/10$ and $n = 4$ – resulting in $\rho_3(x) = 375/(5+x)^4$ – for a run of 50 000 samples, shown in Fig. 1.4.3.

Sampling via transformation of coordinates

2.1 Sampling uniformly within a unit radius disk

Source codes `src/021_disk_sampling.c`

Analysis scripts `src/021_disk_sampling.R`

Executables `exe/021_disk_sampling`

If we want to sample points uniformly distributed over the unit disk, we cannot simply generate $r \sim \mathcal{U}_{[0,1]}$ and $\vartheta \sim \mathcal{U}_{[0,2\pi]}$. The result of doing that is shown in the left graph of Fig. 2.1.1. While the angular distribution poses no problem, there is an undesired higher density of points close to the centre of the disk.

The reason for this becomes clear when you consider the number of points falling inside a ring of given thickness δr . Consider a ring with inner radius r_1 and outer radius $r_1 + \delta r$, with $\delta r \ll r_1$, and another with radii $r_2, r_2 + \delta r$. Despite the ratio of their areas being

$$\frac{\pi[(r_1 + \delta r)^2 - r_1^2]}{\pi[(r_2 + \delta r)^2 - r_2^2]} = \frac{2r_1\delta r + \mathcal{O}(\delta r^2)}{2r_2\delta r + \mathcal{O}(\delta r^2)} = \frac{r_1}{r_2} + \mathcal{O}(\delta r), \quad (2.1.1)$$

the fraction of points within each ring is in both cases proportional to δr , if r is sampled uniformly. Thus, the number of points falling at a distance r should be proportional to r :

$$\rho_r(r) = 2r, \quad (2.1.2)$$

with the 2 in front to ensure normalization over $[0, 1]$. By computing the cumulative and inverting we get

$$F_r(r) = 2 \int_0^r x \, dx = r^2 \implies u = r^2 \implies r = \sqrt{u}. \quad (2.1.3)$$

The graph on the right in Fig. 2.1.1 shows how the sampling becomes visibly uniform with this adjustment.

2.2 Box–Muller transform

Source codes `src/022_box_muller.c`

Analysis scripts `src/022_box_muller.R`

Executables `exe/022_box_muller`

I started from the factorization of $\rho_{x,y}(x, y)$:

$$\rho_{x,y}(x, y) = \frac{1}{2\pi} \exp\left(-\frac{x^2 + y^2}{2}\right) = \rho_x(x)\rho_y(y), \quad \text{with } \rho_x(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}. \quad (2.2.1)$$

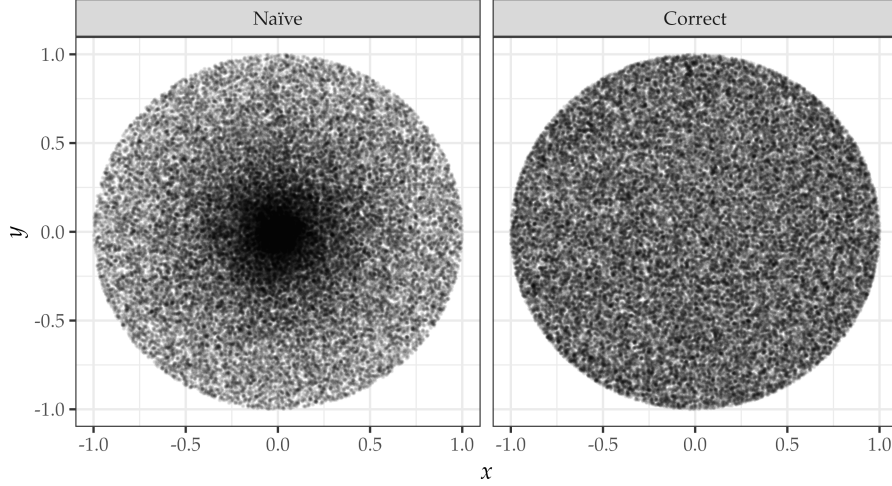


Figure 2.1.1: visualization of 50 000 samples on the unit disk, with the ‘naïve’ method of uniform sampling both on r and ϑ (left), and with the correct method described in the text (right).

If we switch to polar coordinates this becomes, remembering the factor r coming from the Jacobian,

$$\rho_{r,\vartheta}(r, \vartheta) = \frac{r}{2\pi} e^{-r^2/2}, \quad (2.2.2)$$

with $\rho_r(r) = r e^{-r^2/2}$ and $\rho_\vartheta(\vartheta) = 1/2\pi$.

Then, we can marginalize over ϑ to get $\rho_r(r)$:

$$\rho_r(r) = \frac{1}{2\pi} \int_0^{2\pi} r e^{-r^2/2} d\vartheta = r e^{-r^2/2}. \quad (2.2.3)$$

As usual, we compute the cumulative and invert it to generate a random $r \sim \rho_r(r)$:

$$F_r(r) = \int_0^r x e^{-x^2/2} dx = 1 - e^{-r^2/2} \implies e^{-r^2/2} = 1 - u \implies r = \sqrt{-2 \log(1 - u)}. \quad (2.2.4)$$

Then, to get the angle ϑ we recover the *conditional* distribution $\rho_{\vartheta|r}(\vartheta | r)$:

$$\rho_{\vartheta|r}(\vartheta | r) = \frac{\rho_{r,\vartheta}(r, \vartheta)}{\rho_r(r)} = \frac{1}{2\pi}. \quad (2.2.5)$$

Thus, for ϑ we have simply to sample from $\mathcal{U}_{[0,2\pi]}$.

The idea then is to sample two numbers u_1, u_2 from $\mathcal{U}_{[0,1]}$ at each iteration; at that point we can calculate

$$x = \sqrt{-2 \log u_1} \cos(2\pi u_2), \quad y = \sqrt{-2 \log u_1} \sin(2\pi u_2), \quad (2.2.6)$$

to get two numbers x, y distributed according to a standard Gaussian $\mathcal{N}(0, 1)$. We can also get a $z \sim \mathcal{N}(\mu, \sigma^2)$ afterwards by multiplying x or y by σ and summing the mean: $z = \mu + \sigma x$. I ran a simulation with 100 000 samples choosing $\mu = 1$ and $\sigma = 2$, and you can see the results in Fig. 2.2.1.

2.3 Rejection method

Source codes `src/023a_rejection_sampling.c,`
 `src/023b_rejection_accratio.c`

Analysis scripts `src/023_rejection_method.R`

Executables `exe/023a_rejection_sampling,`
 `exe/023b_rejection_accratio`

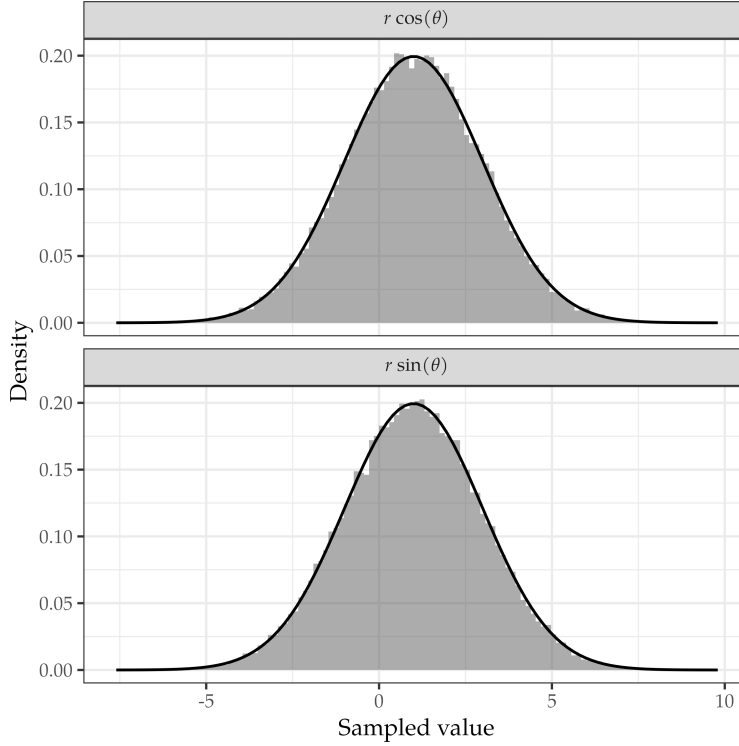


Figure 2.2.1: histograms of 100 000 samples – half in the upper panel, half in the lower one – from a $N(1, 4)$ distribution, obtained with the Box–Muller method. The theoretical distribution is overlaid with a solid line.

Here I implemented the sampling of random numbers from the distribution $f(x) = 2e^{-x^2}/\sqrt{\pi}$ over $[0, \infty)$. We start from a function $g(x)$ that could serve as an upper limit to $f(x)$,

$$g(x) = \begin{cases} A & \text{for } 0 \leq x \leq p, \\ \frac{A}{p} x e^{p^2 - x^2} & \text{for } x > p. \end{cases} \quad (2.3.1)$$

First we normalize it to turn it into a proper probability distribution:

$$1 = \int_0^\infty g(x) dx = Ap + \frac{A}{p} e^{p^2} \int_p^\infty x e^{-x^2} dx = A \left(p + \frac{1}{2p} \right) \implies A = \frac{2p}{1 + 2p^2}. \quad (2.3.2)$$

Then we need a constant c such that $c g(x) \geq f(x)$ everywhere. The maximum of $f(x)$ is in $x = 0$, where $f(0) = 2/\sqrt{\pi}$, so we set $cA = 2/\sqrt{\pi}$. Another thing to consider is that $g(x)$ can have a maximum larger than A in $[p, \infty)$:

$$g'(x) = \begin{cases} 0 & \text{for } 0 \leq x \leq p, \\ \frac{A}{p} (1 - 2x^2) e^{p^2 - x^2} & \text{for } x > p. \end{cases} \quad (2.3.3)$$

Thus in the region $[p, \infty)$ the derivative is zero in $x = 1/\sqrt{2}$. To avoid this we have to choose $p \geq 1/\sqrt{2} \approx 0.707$. An example of sampling with $p = 1/\sqrt{2}$ is given by the histogram in Fig. 2.3.1. A plot of the acceptance ratio as a function of p , starting from the minimum value $1/\sqrt{2}$, is shown in Fig. 2.3.2; obviously the acceptance ratio decreases as p increases, as the gap between $c g(x)$ and $f(x)$ grows wider:

$$\begin{aligned} \int_0^\infty (c g(x) - f(x)) dx &= \frac{2}{\sqrt{\pi}} \int_0^p dx + \frac{2}{p\sqrt{\pi}} \int_p^\infty x e^{p^2 - x^2} dx - 1 \\ &= \frac{2}{\sqrt{\pi}} \left(p + \frac{1}{2p} \right) - 1 \xrightarrow{p \rightarrow \infty} \infty. \end{aligned} \quad (2.3.4)$$

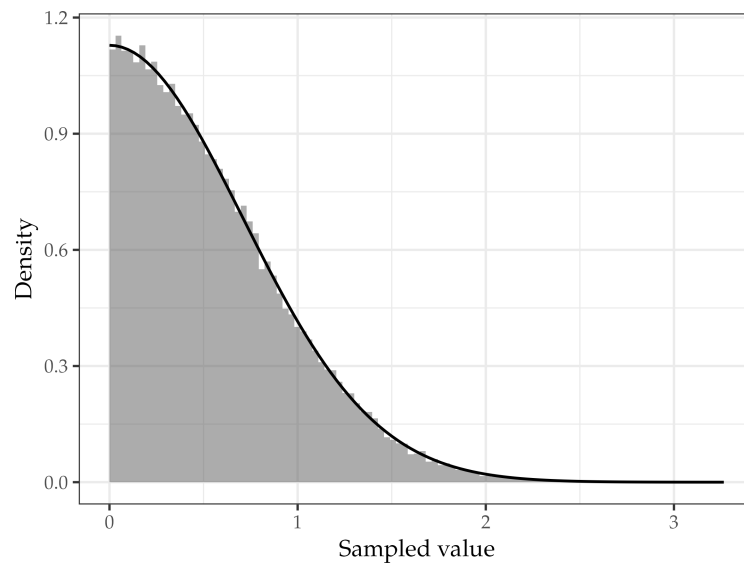


Figure 2.3.1: histogram of 50 000 samples from the distribution $f(x) = 2e^{-x^2}/\sqrt{\pi}$ in $[0, \infty)$ obtained with the rejection method. The theoretical distribution is overlaid with a solid line.

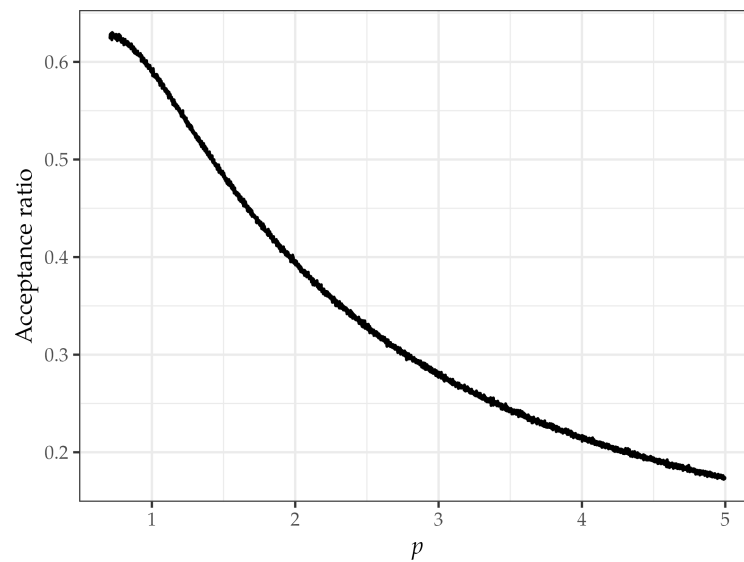


Figure 2.3.2: acceptance ratio over 100 000 samples as a function of p in Ex. 2.3.

Importance sampling

3.1 Comparing importance sampling to the ‘crude’ method

Source codes `src/031_crude_vs_importance.c`

Analysis scripts `src/031_crude_vs_importance.R`

Executables `exe/031_crude_vs_importance`

The goal was to integrate the function $f(x) = g(x)e^{-x^2}$ on $[0, \infty)$, with $g(x)$ being a slowly varying function. I chose $g(x) = x \cos x^2$, so the exact value of the integral is

$$\int_0^\infty f(x) dx = \int_0^\infty x \cos(x^2) e^{-x^2} dx = \dots = \frac{1}{4}. \quad (3.1.1)$$

Following the suggestion, we multiply and divide by a weighting function defined as

$$w(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}, \quad (3.1.2)$$

and so we can write

$$\int_0^\infty g(x) e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \int_0^\infty g(x) w(x) dx = \frac{\sqrt{\pi}}{2} \int_0^\infty g(x) dw(x). \quad (3.1.3)$$

Therefore, we can sample numbers x_i from $w(x)$ and estimate the integral with

$$\int_0^\infty g(x) e^{-x^2} dx \approx \frac{\sqrt{\pi}}{2N} \sum_{i=1}^N g(x_i). \quad (3.1.4)$$

To sample from $w(x)$, which is a half-normal distribution with variance $1/2$, we can use a modified Box-Muller. The idea is to restrict the number generation to the $x > 0$ half-plane. First, since in this case the variance is not unitary, we can recompute the cumulative. The marginal distribution over r needs a factor 2 in front for normalization, and we obtain

$$\rho_r(r) = 2r e^{-r^2} \implies F_r(r) = 1 - e^{-r^2} \implies r = \sqrt{-\log(1 - u_1)}, \quad \text{with } u_1 \sim \mathcal{U}_{[0,1]}. \quad (3.1.5)$$

For ϑ we sample uniformly over $[-\pi/2, +\pi/2]$ in order to stay in the $x > 0$ half-plane:

$$\vartheta = -\frac{\pi}{2} + \pi u_2, \quad \text{with } u_2 \sim \mathcal{U}_{[0,1]}. \quad (3.1.6)$$

Once we have r and ϑ we generate x, y distributed according to $w(x)$ in this way:

$$x = r \cos \vartheta, \quad y = r |\sin \vartheta|. \quad (3.1.7)$$

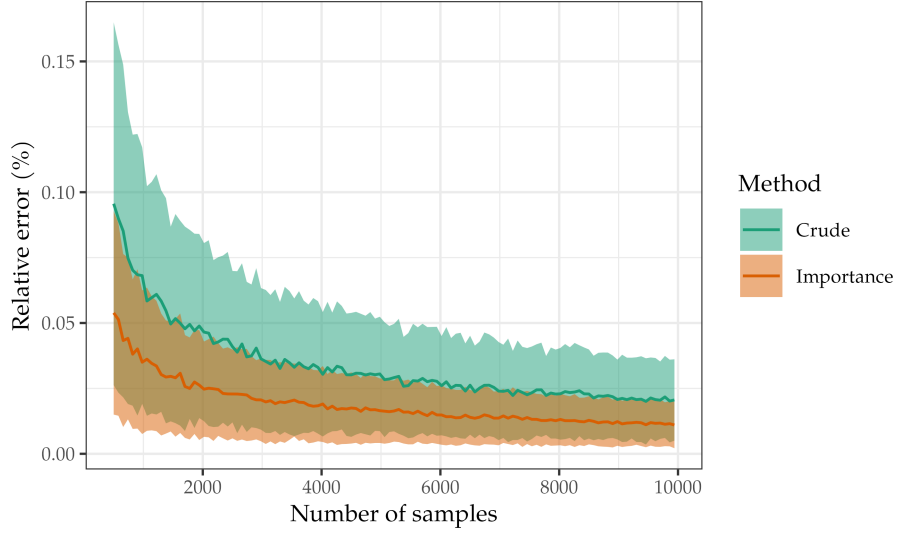


Figure 3.1.1: comparison of the relative error between a ‘crude’ Monte Carlo integration and importance sampling as a function of the number of samples. The solid lines represent the average over 500 realizations, with the standard deviation shaded in the same colour.

In contrast to the importance sampling technique, inside the same C file I also implemented the ‘crude’ Monte Carlo integration, that is a simple average of a set of $f(x_i)$ with x_i sampled uniformly over an interval from 0 to a given $x_{\max}$:

$$\int_0^\infty g(x)e^{-x^2} dx \approx \frac{1}{N} \sum_{i=1}^N g(x_i)e^{-x_i^2}, \quad \text{with } x_i \sim \mathcal{U}_{[0, x_{\max}]}. \quad (3.1.8)$$

You can see the comparison between the two methods in terms of relative error in Fig. 3.1.1. The importance sampling is definitely more efficient, although the crude method has a higher variance which can lead to a lower error in lucky runs. Also, as the number of samples grows the difference between the two methods shrinks, but this is rather unsurprising.

3.2 Optimizing importance sampling for cosine integration

Source codes `src/032_cosx_importance.c`

Analysis scripts `src/032_cox_importance.R`

Executables `exe/032_cosx_importance`

This exercise asked to compute the integral

$$\int_0^{\pi/2} \cos x \, dx \quad (3.2.1)$$

using once again the importance sampling technique, with a weighting function $g(x) = a + bx^2$, a and b to tune. To have the best performance, the idea is to have a $g(x)$ that is large where the integrand is large and small when the integrand is small. First, $g(x)$ should be normalized, which brings down the number of parameters to one:

$$1 = \int_0^{\pi/2} (a + bx^2) dx = \frac{\pi}{2}a + \frac{\pi^3}{24}b \implies a = \frac{2}{\pi} - \frac{\pi^2}{12}b. \quad (3.2.2)$$

Next, since we want $g(x)$ to be as similar as possible to $\cos x$, we can restrict ourselves to $b < 0$. If b is too negative, though, $g(x)$ becomes negative in part of the domain. The best solution is to impose $g(\pi/2) = 0$:

$$0 = \frac{2}{\pi} - \frac{\pi^2}{12}b + \frac{\pi^2}{4}b \implies b = -\frac{12}{\pi^3} \implies g(x) = \frac{3}{\pi} \left(1 - \frac{4}{\pi^2}x^2 \right). \quad (3.2.3)$$

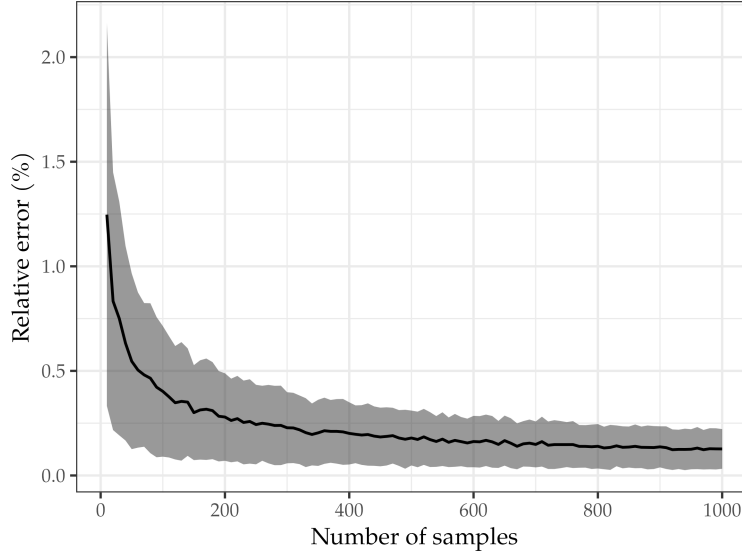


Figure 3.2.1: relative error in the integration of $\cos x$ between 0 and $\pi/2$ with the importance sampling technique, as described in Ex. 3.2. The solid lines represent the average over 500 realizations, with the standard deviation shaded in grey.

To sample from g , we compute the cumulative as usual:

$$G(x) = \frac{3}{\pi} \int_0^x \left(1 - \frac{4}{\pi^2} y^2\right) dy = \frac{3}{\pi} x - \frac{4}{\pi^3} x^3. \quad (3.2.4)$$

The inversion requires solving a cubic equation,

$$u = \frac{3}{\pi} x - \frac{4}{\pi^3} x^3 \implies x^3 - \frac{3}{4} \pi^2 x + \frac{\pi^3}{4} u = 0. \quad (3.2.5)$$

Let's define $\alpha = \pi^2/4$:

$$x^3 - 3\alpha x + \pi\alpha u = 0. \quad (3.2.6)$$

This is a depressed cubic, so thankfully it has a trigonometric solution given by¹

$$x_k = 2\sqrt{\alpha} \cos \left[\frac{1}{3} \arccos \left(-\frac{3\pi\alpha u}{6\alpha} \sqrt{\frac{\beta}{\beta\alpha}} \right) - \frac{2\pi}{3} k \right], \quad \text{with } k = 0, 1, 2. \quad (3.2.7)$$

With some algebra, and selecting the right k , we can arrive at the much simpler form

$$x = \pi \sin \left(\frac{1}{3} \arcsin u \right), \quad \text{with } u \sim \mathcal{U}_{[0,1]}. \quad (3.2.8)$$

To wrap up, what I ended up doing is

$$\int_0^{\pi/2} \cos x \, dx \approx \frac{1}{N} \sum_{i=1}^N \frac{\cos x_i}{g(x_i)} = \frac{\pi}{3N} \sum_{i=1}^N \frac{\cos x_i}{1 - 4x_i^2/\pi^2}, \quad (3.2.9)$$

with each x_i generated according to Eq. (3.2.8). The integration error as a function of the number of iterations is reported in Fig. 3.2.1. Remarkably as low as 100 or 200 iterations are sufficient to get, on average, an error that is comfortably under 1 %.

1. https://en.wikipedia.org/wiki/Cubic_equation#Trigonometric_and_hyperbolic_solutions.

3.3 Analysing the variance with or without importance sampling

Analysis scripts src/033_variance_comparison.R

This exercise asked to investigate the improvement one can get with importance sampling applied to a simple integral, the average on the distribution $\rho(x) = e^{-x}$, $x \geq 0$, of the function

$$f(x) = \begin{cases} 0 & \text{for } 0 \leq x < T, \\ 1 & \text{for } x \geq T. \end{cases} \quad (3.3.1)$$

This average can be readily calculated as

$$\langle f \rangle_\rho = \int_0^\infty f(x) e^{-x} dx = \int_T^\infty e^{-x} dx = e^{-T}. \quad (3.3.2)$$

The problem of this integral is that $f(x)$ is different from 0 in the region where $\rho(x)$ is quickly vanishing. This means that if we were to sample numbers from $\rho(x)$ we would waste a lot of iterations.

Consider instead the function $g(x; a) = ae^{-ax}$ defined for $x \geq 0$ and $a \in (0, 1]$. We can use it for an importance sampling and optimize a to minimize the variance. In particular, defining $F(x) := f(x)\rho(x)/g(x)$ we can write

$$\langle f \rangle_\rho = \int_0^\infty f(x)\rho(x) dx = \int_0^\infty \frac{f(x)\rho(x)}{g(x)} g(x) dx = \langle F \rangle_g. \quad (3.3.3)$$

Let's calculate the second moment now:

$$\begin{aligned} \langle F^2 \rangle_g &= \int_0^\infty \frac{f^2(x)\rho^2(x)}{g^2(x)} g(x) dx = \int_T^\infty \frac{\rho^2(x)}{g(x)} dx \\ &= \frac{1}{a} \int_T^\infty e^{-(2-a)x} dx = \frac{e^{-(2-a)T}}{a(2-a)}. \end{aligned} \quad (3.3.4)$$

By subtracting the first moment squared we get the variance,

$$\sigma_g^2(F) = \langle F^2 \rangle_g - \langle F \rangle_g^2 = \frac{e^{-(2-a)T}}{a(2-a)} - e^{-2T}. \quad (3.3.5)$$

The variance of f with respect to ρ is given instead by

$$\begin{aligned} \sigma_\rho^2(f) &= \int_0^\infty (f(x) - \langle f \rangle_\rho)^2 \rho(x) dx = \int_T^\infty \rho(x) dx + \langle f \rangle_\rho^2 - 2\langle f \rangle_\rho \int_T^\infty \rho(x) dx \\ &= \langle f \rangle_\rho + \langle f \rangle_\rho^2 - \langle f \rangle_\rho^2 \\ &= e^{-T}(1 - e^{-T}). \end{aligned} \quad (3.3.6)$$

Now, let's minimize $\sigma_g^2(F)$ with respect to a :

$$\frac{\partial}{\partial a} \sigma_g^2(F) = e^{-(2-a)T} \left[\frac{Ta(2-a) - 2(1-a)}{a^2(2-a)^2} \right], \quad (3.3.7)$$

which gets us to

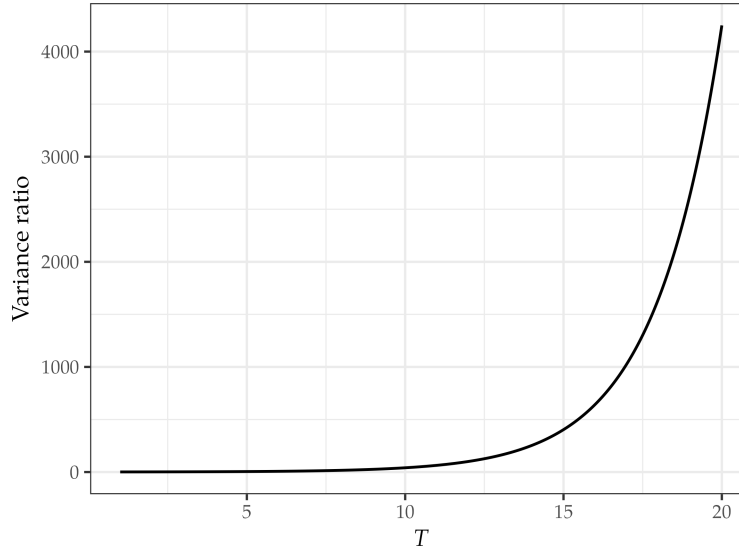
$$Ta^2 - 2(1+T)a + 2 = 0 \iff a = a_\pm^* = \frac{1+T \pm \sqrt{1+T^2}}{T}. \quad (3.3.8)$$

Only a_-^* is compatible with the condition $a \in (0, 1]$, so the optimal a is given by

$$a^* = \frac{1+T - \sqrt{1+T^2}}{T}. \quad (3.3.9)$$

Table 3.3.1: variance ratios for some values of T , all evaluated at a^* .

T	$\sigma_\rho(f)/\langle f \rangle_\rho$	$\sigma_g(F)/\langle F \rangle_g$	$\sigma_\rho(f)/\sigma_g(F)$
3	4.37	1.95	2.24
5	12.14	2.55	4.76
10	148.41	3.65	40.71
20	22026.47	5.18	4249.17

**Figure 3.3.1:** variance ratio $\frac{\sigma_\rho(f)}{\sigma_g(F)}$ as a function of T , from Ex. 3.3.

In Tab. 3.3.1 I calculated the ratios between each of the two variances and $\langle f \rangle_\rho$ and the ratio between the two variances, for some values of T . While $\sigma_g(F)/\langle f \rangle_\rho$ grows slowly, $\sigma_\rho(f)/\langle f \rangle_\rho$ has an exponential increase; indeed

$$\frac{\sigma_\rho(f)}{\langle f \rangle_\rho} = \frac{\sqrt{e^{-T}(1 - e^{-T})}}{e^{-T}} = \sqrt{e^T - 1}. \quad (3.3.10)$$

The variance ratio as function of T is also plotted in Fig. 3.3.1.